

Short Paper: What a disclosed reserve impairment cannot explain in the USDC de-peg

Anonymous Submission to FC 2027

Abstract. When an issuer publicly quantifies how much of its reserve is impaired while the outcome is unresolved, that disclosure pins a worst-case floor under the coin’s value. We turn that into a reusable public-data instrument needing three preconditions: the impairment is quantified while unresolved, the price falls materially below par while the disclosure is live, and the fundamental channel is reserve-only. Where the observed low breaches the floor, it returns a lower bound on the part of the de-peg that expected loss cannot account for. Otherwise it returns no verdict. Screening candidate episodes leaves one qualifying application, the March 2023 USD Coin de-peg, where the observed low does breach the floor. A cross-coin placebo and three-episode panel sit outside the estimator. Because the residual covers friction as well as panic, it speaks to backstop and redemption design rather than reserve-quality rules alone.

Keywords: Stablecoins · Bank runs · De-peg · Financial stability · Market microstructure

1 Introduction

Stablecoins are becoming payment infrastructure, and the GENIUS Act gives them their first comprehensive U.S. federal framework [36]. A fiat-backed stablecoin that trades below par admits two readings. Under rational repricing, an impaired reserve is worth less. Under non-fundamental repricing, a self-fulfilling coordination run [14,20] and microstructure friction push the price below the floor a reserve loss can justify. No measurement of March 2023 cited in Section 2 forms that floor. A de-peg with a non-fundamental share bounded away from zero calls for backstop and redemption design, since a reserve-quality rule addresses only the fundamental share.

We offer a reusable estimator for any disclosed-impairment de-peg. From an issuer’s disclosed reserve impairment, derive a worst-case floor. Test whether the observed trough breaches that floor. Read the breach, if any, as a bound on the non-fundamental share. Figure 1 shows those three steps. A supervisor can run it on any public hourly print, within hours of a break. It returns no verdict when the trough does not breach the floor.

We apply it to the March 2023 USD Coin (USDC) de-peg, where Silicon Valley Bank (SVB) failed holding about 8% of reserves, and a joint federal backstop [5] restored the peg within days.

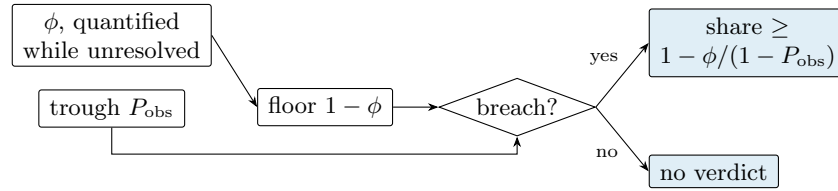


Fig. 1. The estimator. A disclosed impairment gives a floor, and a public price that breaches it bounds the non-fundamental share.

Under expected-loss pricing, fundamental value cannot fall below $1 - \phi$, where ϕ is the disclosed impairment fraction. Under (A4), a preference-free corollary extends that floor to any monotone-utility valuation. The observed trough breaches the floor at every admissible unremediated-loss probability and recovery rate, and requires $q^* = 1.54$, which exceeds one. Under (A1)–(A3), no expected-loss account of the disclosed impairment is consistent with the trough. At the disclosed ϕ the lower edge of the resulting non-fundamental share runs from 19% on Kraken’s close to 35% on the composite. It stays positive across impairment readings, and at the full re-peg is 100% ex post, an identity (Section 4).

The paper makes two contributions:

- **An identification argument** (Proposition 1) turning a disclosed impairment into a floor that a price either breaches or does not, with the breach read as a bound on the non-fundamental share.
- **A screen** over six candidate de-pegs, none of which satisfies all three pre-conditions, so the single qualifying application is a property of the disclosure record rather than of the method.

Two analyses stay outside the estimator: the ex-ante fragility ordering by reserve exposure that the realized troughs match, with null probability $1/2$ (Section 5), and a three-episode panel (Section 6).

2 Related Work

Since 2020 a literature applies run theory to stablecoin design and peg arbitrage [30,23,27,2]. Anadu et al. [3] analyze the March 2023 de-peg through a discrete “break-the-buck” threshold. Watsky et al. [38], and a later Federal Reserve post-mortem [16], attribute that break to a redemption window restricted to Circle’s approved customers over the March 10–13 weekend. Secondary prices de-peged while the arbitrage loop was closed. Uhlig [35] finds many TerraUSD holders behaved as rational Bayesian updaters in May 2022, evidence that rational and panic behavior are hard to separate without a structural restriction.

Other work measures the March 2023 de-peg without an expected-loss floor: the SVB link [19], a daily cross-coin exposure ranking [15], a detection lead of about five hours before USDC crosses \$0.99 [7], a settlement-friction framing [25],

and the on-chain contagion pathway [33]. Our bound derives an expected loss from disclosed exposure. None of the papers just cited does.

Contribution boundary. Structural calibrations [28] estimate a run probability, and event studies [3] document flows. We bound one episode’s realized non-fundamental share from an accounting identity. Anadu et al.’s [3] rational run concerns the redemption decision and makes no claim about whether the price was justified. Watsky et al.’s [38] weekend discount is the microstructure friction the non-fundamental share includes. Ahmed, Aldasoro, and Duley [1] run synthetic-control tests on four other episodes and treat March 2023 only as a price path. Zhu [39] classifies which channel moves price and leaves how far unmeasured. Molinet and Filos-Ratsikas [31] define fiat-backed coins as the always-stable case ($1/p_t = 1$) and never analyze March 2023. Eichengreen, Nguyen, and Viswanath-Natraj [18] back a devaluation probability out of stablecoin prices, where the probability ours implies exceeds one.

3 Setting and Data

USDC is a fiat-backed stablecoin redeemable one-for-one against Circle’s reserves of short-dated Treasuries and bank cash, so a bank failure impairs backing through the cash leg and through the arbitrage loop of redemption at par.

Silicon Valley Bank failed on March 10, 2023. Late that evening, Circle disclosed on its official X account that “\$3.3 billion of the ~\$40 billion of USDC reserves remain at SVB” [8]. Circle restated that figure as “about 8%” of total reserves in its March 12 press release [9], and contemporaneous reporting corroborates it [6]. We take $\phi \approx 0.08$, Circle’s own rounded disclosure. The secondary-market price fell below par over March 10–11, reaching a trough of \$0.877. On March 12 at 22:15 UTC, the FDIC, the Federal Reserve, and the Treasury issued a joint statement guaranteeing SVB depositors in full [5], and USDC restored its peg over the following days.

DAI, the contagion coin, is linked to USDC through MakerDAO’s peg-stability module, which swaps the two one-for-one up to a debt ceiling that press reporting says was reached on 11 March 2023 [12]. Tether (USDT), the largest fiat-backed stablecoin, had no disclosed exposure to SVB and serves as the unexposed coin.

All analysis runs offline on a frozen snapshot over the window 2023-03-08 to 2023-03-17 (UTC), and no proprietary data enters it. The headline USD price series is an hourly cross-venue aggregate from the public aggregator DeFiLlama [13]. The frozen record does not enumerate its constituent venues, so we do not treat it as an independent venue. Kraken’s fiat-settled USDC/USD close and VWAP [24] cross-check the \$0.877 trough, at minimum prints of \$0.901 and \$0.893. The pair’s \$0.874 hourly low rests on a single 3,373-USDC trade, so we do not read the cross-check off it. In Figure 2(b) circles mark Kraken’s close, drawn at its hour’s end, and triangles its VWAP, drawn at the midpoint.

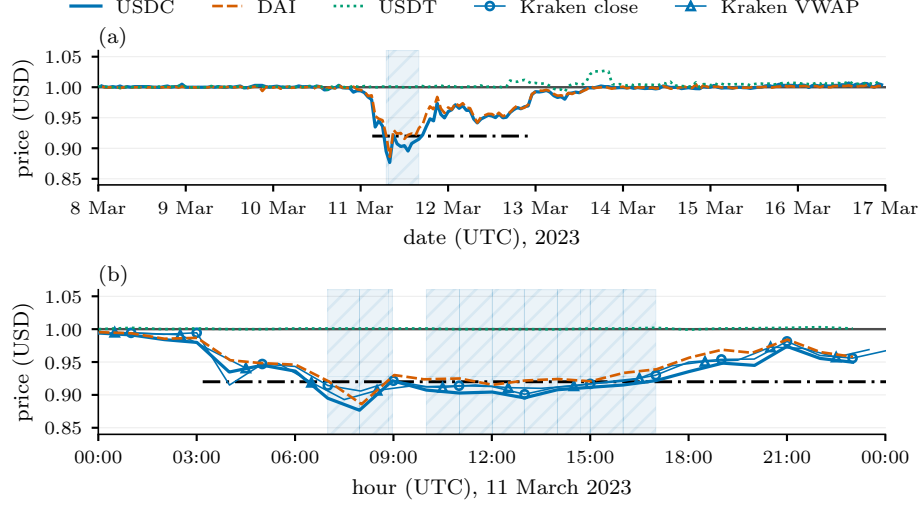


Fig. 2. Hourly USD prices of USDC, DAI and USDT over the window (a) and over 11 March 2023 (b), against par and the \$0.92 floor while the disclosure was live.

4 The Bound and Its Application

The estimator is pure accounting, with no run threshold and no fitted parameter. Let ϕ be the fraction of reserves impaired, $\rho \in [0, 1]$ the expected recovery rate on the impaired assets, and $q \in [0, 1]$ the probability under the pricing measure that the loss on the impaired fraction goes unremediated. Expected backing per coin is

$$P_{\text{fund}}(q, \rho) = 1 - q\phi(1 - \rho), \quad (1)$$

and its minimum over q and ρ , the worst-case floor, is $1 - \phi$. The estimator rests on three assumptions, and a fourth serves the preference-free argument below.

- (A1) **Reserve-only fundamental.** A coin's fundamental value is its expected reserve backing per unit given disclosed reserves, and the only impairment is the exposure ϕ .
- (A2) **Par redemption when solvent.** Conditional on rescue, redemption is at \$1 within the frozen window.
- (A3) **Bounded recovery.** Recovery on impaired reserves satisfies $\rho \in [0, 1]$, because the coin is fully reserved and the reserves carry no leverage.
- (A4) **Monotone-utility valuation.** The unconstrained marginal holder values the claim as an expected utility of its payoff, for some monotone increasing utility and some probability measure over outcomes, with no ambiguity about the disclosed ϕ . This nests risk neutrality, any degree of risk aversion, and rank-dependent weighting.

Proposition 1 (Rational-pricing floor and impossibility). *Assume (A1)–(A3). Then $P_{\text{fund}}(q, \rho) \in [1 - \phi, 1]$ for all $q, \rho \in [0, 1]$, so no expected loss*

on the disclosed impairment justifies a value below $1 - \phi$. The unremediated-loss probability an observed price $P_{\text{obs}} < 1 - \phi$ would require, at the worst-case recovery $\rho = 0$, is $q^* = (1 - P_{\text{obs}})/\phi > 1$.

Proof. P_{fund} is decreasing in q and increasing in ρ , so it is minimized at $(q, \rho) = (1, 0)$ at value $1 - \phi$ and maximized at $q = 0$ at value 1, and q^* solves $P_{\text{obs}} = 1 - q\phi$ at $\rho = 0$. $\square$

This is the floor-at-recovery-value bound of credit pricing [29,17].

Under (A1)–(A4) the floor is also preference-free. By (A2) and (A3) the payoff X on a USDC claim has support in $[1 - \phi, 1]$ in every state. For any monotone increasing utility u , the certainty equivalent $u^{-1}(\mathbb{E}[u(X)])$ of a lottery with that support lies in $[1 - \phi, 1]$ as well. That floor bounds rational valuations, meaning willingness to pay. It does not bound market-clearing prices, because a seller under binding margin or capital constraints can rationally transact below every unconstrained valuation [34,22]. That channel is a friction, and the non-fundamental share defined next includes it. The floor is only as strong as the support condition $X \in [1 - \phi, 1]$, which does not cover issuer-wrapper risk [21].

Write the observed deviation $1 - P_{\text{obs}}$ and the fundamental deviation $\phi(1 - \rho)$. The non-fundamental share of the de-peg is

$$s_{\text{nf}}(\rho) = \frac{(1 - P_{\text{obs}}) - \phi(1 - \rho)}{1 - P_{\text{obs}}}. \quad (2)$$

Its lower edge takes $\rho = 0$, treating total permanent loss of the impaired reserves as the fundamental cap. Its upper edge takes $\rho = 1$, the full re-peg, which implies zero permanent loss and gives $s_{\text{nf}}(1) = 1$, or 100% ex post, an identity. The residual covers coordination panic and microstructure friction together, so the share is an upper bound on pure coordination panic.

With the disclosed impairment $\phi = 0.08$, the worst-case floor is $1 - \phi = \$0.92$ and the observed trough is $P_{\text{obs}} = \$0.877$.¹ The trough lies below the floor, so under (A1)–(A3) no expected loss rationalizes it, and the implied unremediated-loss probability is $q^* = (1 - 0.877)/0.08 = 1.54$. The lower edge of (2) is $(0.1233 - 0.08)/0.1233 = 0.351$, so at least 35% of the de-peg is non-fundamental. Across three readings of Circle’s disclosure and five trough readings across four feeds, the fifth being the composite’s second-lowest print, the lower edge runs from 16.7% to 36.4%, and every one of those fifteen combinations breaches the floor. At the disclosed ϕ , the lowest is 19.2%, from Kraken’s close, whose minimum print is $\$0.901$. By Proposition 1 the trough is unrationalizable for any impairment below 12.3%, a 4.3-point margin over the disclosed 8%, and the lower edge stays strictly positive across that margin. Figure 3 plots that edge against ϕ for the composite, Kraken close and VWAP troughs, each crossing zero at the impairment that would make its own trough rationalizable.

¹ Prose rounds troughs to three figures. Computation uses the frozen four-figure print, so $\$0.877$ enters as 0.8767 and Section 5’s deviations reconcile at full precision.

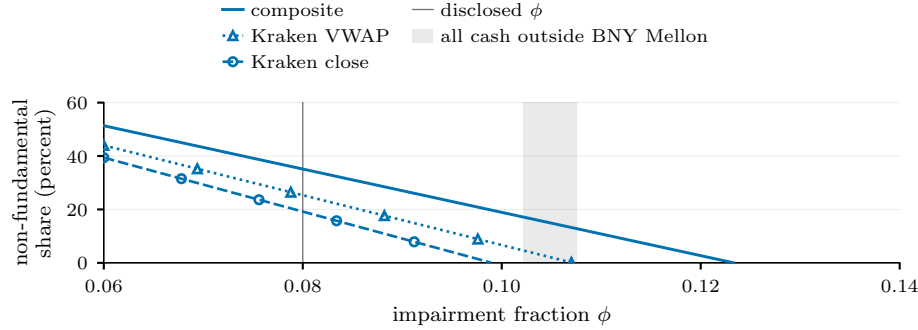


Fig. 3. Lower-edge non-fundamental share against impairment. Curves cross zero at one minus their troughs. Shading treats cash outside BNY Mellon as impaired.

Under (A1), ϕ is Circle’s disclosed figure, and evaluating at $\rho = 0$ uses the lowest floor (A3) allows. Against the circulating supply at the trough, which exceeds the \$40B figure Circle used, ϕ is 0.079, below the headline 0.08.

Circle’s disclosure quantifies only the SVB deposit, but the same update itemizes the whole cash leg: \$9.7B, of which \$5.4B sat at BNY Mellon, \$3.3B at SVB and \$1B at Customers Bank, with no exposure to Silvergate [10]. With the \$32.4B Treasury leg the same update reports, reserves total \$42.1B. Treating all cash outside BNY Mellon as impaired gives $\phi = 0.102$ on that total, where the lower edge is 17.2% on the composite and 4.7% on Kraken’s VWAP and Kraken’s close returns no verdict. On the \$40B base the same reading gives $\phi = 0.1075$, where only the composite still returns a verdict, at 12.8%.

Kraken’s 240 hourly bars run one day longer than the composite’s 216-hour window, in which four snapshots are absent, and against that denominator its close breaches the \$0.92 floor in 9 hours, its VWAP in 10, and its hourly low in 12. Sub-floor trades themselves fall in twelve clock hours, 37,303 of them worth \$154.0M.

Because the estimator needs only a price and a publicly quantified ϕ , it can be evaluated at every hour of the frozen window (Figure 2). Of USDC’s 212 in-window hourly readings, 9 lie below the \$0.92 floor, and all nine fall on 11 March 2023 between 06:59 and 16:00 UTC, shaded in both panels. The tenth hour of that window, at 08:57, recovers to \$0.9216. That impossibility window measures duration and specificity. It adds no evidence volume, because hourly prices within one episode are near-unit-root, with lag-1 autocorrelation $r = 0.963$, so the 212 readings carry an AR(1)-adjusted effective sample size $n(1-r)/(1+r) = 4.0$. The lower edge, q^* , and the floor-breach indicator are three readings of $1 - P_{\text{obs}} - \phi$, so their agreement corroborates nothing.

5 Cross-Coin Evidence

Reserve exposure orders the three coins ex ante: USDC held direct SVB exposure, DAI was pinned to USDC through the peg-stability module, and USDT held none. That gives the ex-ante fragility ordering USDC, DAI, USDT. Troughs are \$0.877, \$0.886, and \$0.992 in that order. The deviations from par are 1233, 1141, and 76 bps. DAI’s mechanical link to USDC leaves one free comparison, so the realized ordering has probability $1/2$ and illustrates the ex-ante ordering without testing it. USDC and DAI clear the 1% break threshold, a deviation of at least one cent from par, by factors of 12 and 11. USDT does not clear it, so the placebo criterion is satisfied: the exposed coins break and the unexposed coin does not.

USDT as the unexposed control. USDT was bid to a premium while USDC and DAI fell. That pattern is consistent with the crypto safe-haven behavior documented for USDT [4]. The premium is venue- and time-specific. On Coinbase USDT-USD [11] it peaked at +266 bps at 02:00 UTC on 11 March, 71 minutes before Circle’s 03:11 UTC disclosure. Over the window \$1.24B of \$1.52B traded above par, and that peak is one hour of it. The composite feed peaks later and higher, at +270 bps at 19:00 UTC on 13 March. That peak falls after the joint federal backstop of the previous evening, so it is post-backstop rather than intra-run. The Coinbase premium responds to the SVB failure itself, because Circle’s USDC-specific disclosure had not yet been made. A common-factor repricing of stablecoin risk would move all three coins down together, so it cannot produce that premium. Rational substitution out of a gated claim can, and the two are not separated here.

6 Screening

The estimator requires three preconditions. First (i), ϕ must be publicly quantified while the exposure is unresolved. Second (ii), the price must fall more than 1% below par while that disclosure is live. Third (iii), the fundamental channel must be reserve-only, so that $1 - \phi$ is the floor. Beyond the application itself and Tether (2019), we screened six candidate episodes: BUSD, Tether (2022), USDR, USDP, GUSD, and TUSD. None satisfies all three, and each fails in one of two ways: no disclosed impairment, or a loss measured under a different valuation model. Issuers with a material impairment did not disclose its size in any of the six, so one qualifying episode is a property of the disclosure record.

Three episodes bound the estimator’s behavior: the qualifying application in USDC (2023), and two that return no verdict, Tether (2019) and Tether (2022), which is USDT over the Terra episode of May 2022 [26]. USDC fires on Circle’s 03:11 UTC disclosure of 11 March [8]: $\phi = 0.08$, floor \$0.92, composite headline trough \$0.877 at 07:57 UTC, and $q^* = 1.54$. Tether’s episode is assessed under precondition (i). The New York Attorney General’s 25 April 2019 statement [32] quantified Tether’s exposure and Tether contested the characterization. The statement implies $\phi \in [0.2585, 0.3324]$ and a floor in $[0.6676, 0.7415]$.

The deepest post-disclosure Kraken USDT/USD print, \$0.9551 on 26 April 2019, is 21.4 to 28.8 cents above every worst-case floor, and $q^* \in [0.135, 0.174]$, so the verdict is no verdict, above floor. We read both episodes at zero recovery on the impaired reserves, so the same one-sided bound governs both. Tether (2022) has no disclosure, so $\phi = 0$ and precondition (i) fails. The floor would be par, which any sub-par print breaches as an identity and not as evidence, so the verdict is no verdict, undisclosed.

Read strictly, precondition (i) demands the issuer’s own public quantification while the exposure is unresolved, and under that reading the panel contains no validated true negative. One episode on which the bound returns no verdict does not measure a false-positive rate. The specificity of the bound is unmeasured, and unmeasurable on the current record, because a validated true negative needs an issuer-disclosed ϕ whose trough stays above $1 - \phi$, and no screened episode supplies one.

TrueUSD’s unremediated impairment during our window was not publicly quantified until a September 2024 SEC complaint [37], so precondition (i) fails and the verdict is again no verdict, undisclosed. The residual is defined relative to what the market could know, because a concealed impairment produces no signal.

7 Limitations and Conclusion

The panic-versus-fundamentals split is contested [3,38]. The composite feed could mismeasure the trough. The disclosed ϕ could understate exposure. The breakeven impairment is 12.3% on the composite and 9.9% on Kraken’s close, and the worst case the disclosure admits, $\phi = 0.102$, sits between them. A time-value discount prices a delayed redemption for a holder who has an approved claim. Convertibility risk is the cost to a holder without an approved claim during the restricted window. That cost is not bounded by the discount and belongs to the residual. No public data source can bound it further. We identify the mechanism from a single episode. The ex-post 100% is an aggregate statement, and holders who sold at the trough still realized losses.

The frozen snapshot, the seeded analysis code, and a manifest with a SHA-256 digest for every file it lists sit in an anonymized artifact at <https://anonymous.4open.science/r/usdc-depeg-artifact-fc27>. Every computed number and figure regenerates offline from the artifact without an API key.

A disclosed impairment implies a worst-case floor, and the USDC trough lies below it. Under (A1)–(A3) and at the disclosed ϕ , at least 35% of the de-peg was non-fundamental by impossibility on the composite feed, and at least 19% on Kraken’s close, the most conservative feed. At the worst case the disclosure itself admits, $\phi = 0.102$, the composite still returns a verdict and Kraken’s close does not. Whether that residual is coordination panic or convertibility friction on a closed redemption rail, both point to backstop and redemption design rather than reserve-quality rules alone, so the policy reading does not depend on the split. The estimator needs only a price and a publicly quantified ϕ .

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